

Life Processes in Plants

Science — Biology · Class 7 · Max Marks: 25 · Time: 45 min

Section A — Multiple Choice Questions ($1 \times 8 = 8$)

1. The process by which green plants make their own food is:

- A. transpiration
- B. respiration
- C. digestion
- D. photosynthesis

2. The raw materials needed for photosynthesis are:

- A. nitrogen and water
- B. carbon dioxide and water
- C. oxygen and water
- D. oxygen and glucose

3. Photosynthesis takes place mainly in the ____ of a leaf, which contain chlorophyll:

- A. green cells
- B. roots
- C. woody stem
- D. flowers

4. The tiny pores on a leaf through which gases are exchanged are called:

- A. xylem
- B. villi
- C. stomata
- D. veins

5. *Cuscuta* (amarbel) has no green leaves and grows on other plants, taking food from them. Its nutrition is:

- A. saprotrophic
- B. autotrophic
- C. insectivorous
- D. parasitic

6. The pitcher plant traps and digests insects mainly to obtain:

- A. nitrogen it cannot get from the soil
- B. sunlight
- C. carbon dioxide
- D. water

7. Water and minerals absorbed by the roots are carried upward through tubes called:

- A. stomata
- B. phloem
- C. xylem
- D. veins

8. The loss of water as vapour from the leaves of a plant is called:

- A. respiration
- B. transpiration
- C. absorption
- D. photosynthesis

Section B — Fill in the Blanks ($1 \times 5 = 5$)

1. The green pigment in leaves that traps sunlight is called ____.

2. During photosynthesis, plants give out the gas ____.

3. Food made in the leaves is carried to the rest of the plant through the ____.

4. Fungi that obtain food from dead and decaying matter show ____ nutrition.

5. The alga and the fungus in a lichen live together for mutual benefit; this relationship is called ____.

Section C — True or False ($1 \times 5 = 5$)

1. Photosynthesis can take place only when sunlight is available. (True / False)
2. Plants respire only at night. (True / False)
3. Xylem carries food from the leaves to the rest of the plant. (True / False)
4. Insectivorous plants are non-green and cannot carry out photosynthesis. (True / False)
5. Transpiration helps to pull water upward from the roots to the leaves. (True / False)

Section D — Match the Columns (1 × 4 = 4)

Column A	Column B
1. Chlorophyll	a) transport of food
2. Stomata	b) traps sunlight
3. Xylem	c) exchange of gases in the leaf
4. Phloem	d) transport of water and minerals

Section E — Assertion–Reason (1 × 3 = 3)

Choose: (A) both A and R true and R explains A; (B) both true but R does not explain A; (C) A true, R false; (D) A false, R true.

1. **Assertion (A):** Leaves are called the food factories of a plant.
Reason (R): Photosynthesis takes place mainly in the leaves.
2. **Assertion (A):** Insectivorous plants trap and digest insects.
Reason (R): They grow in nitrogen-poor soil and obtain nitrogen from the insects.
3. **Assertion (A):** Desert plants such as cactus have very few, spiny leaves.
Reason (R): This increases the loss of water by transpiration.

Answer Key

Multiple Choice Questions

- | | | | | |
|------|------|------|------|------|
| 1. D | 2. B | 3. A | 4. C | 5. D |
| 6. A | 7. C | 8. B | | |

Fill in the Blanks

1. chlorophyll
2. oxygen
3. phloem
4. saprotrophic
5. symbiosis (symbiotic nutrition)

True or False

1. True
2. False
3. False
4. False
5. True

Match the Columns

1-b, 2-c, 3-d, 4-a

Assertion–Reason

1. A
2. A
3. C